Techfest IIT Bombay 2026–27 | Deliverable 4

Namma Space: Code Repository & Setup Instructions

Complete Guide for Evaluators to Reproduce 3D Reconstruction & Web Roaming

1. System Prerequisites

- **Operating System:** Windows 10/11 or Ubuntu Linux 22.04 LTS.
- **Python:** Python 3.10 with PyTorch 2.1+ and CUDA 11.8 / 12.1.
- **Node.js:** Node.js 18+ and npm for the frontend viewer.
- **GPU:** NVIDIA GPU (e.g., RTX 3060, 4060, or higher) with ≥ 6 GB VRAM.
- **Tracking Tools:** COLMAP 3.8+ (included/discovered) OR Jawset Postshot CLI.

2. One-Time Environment Setup

```
# 1. Clone repository
git clone https://github.com/namma-space/namma_space.git
cd namma_space/namma_space

# 2. Setup Python environment
pip install -r requirements.txt
pip install -r reconstruction/requirements.txt
pip install -r backend/requirements.txt

# 3. Setup Frontend dependencies
cd frontend
npm install
cd ..
```

3. Single-Command End-to-End Reconstruction

To reconstruct any smartphone video (e.g. `EX3.MOV` or test sample data):

```
cd reconstruction

# Using Built-in Balanced Pipeline (10,000 iters, 12 P-core threads):
python run_pipeline.py --video "EX3.MOV" --clean

# OR Using Fast GPU Postshot Acceleration:
python run_pipeline.py --video "EX3.MOV" --postshot --clean
```

> **What the Pipeline Executes Automatically:**
>
> 1. Extracts keyframes at 2.5 FPS with Laplacian blur filtering.
> 2. Estimates camera poses via incremental bundle adjustment / GPU tracking.
> 3. Aligns the room geometry to the horizontal plane using RANSAC.
> 4. Optimizes 3D Gaussian Splats with 3rd-order spherical harmonics.
> 5. Builds the 5 cm obstacle occupancy grid and registers the scene in SQLite.
> 6. Exports `.splat` and `.glb` directly to `backend/static/models/`.

4. Launching the Web Viewer

```
# Terminal 1: Start FastAPI Backend Server
cd backend
python -m uvicorn main:app --host 127.0.0.1 --port 8000

# Terminal 2: Start Vite Frontend Web App
cd frontend
npm run dev
```

Open `http://localhost:5173/` in any web browser to explore your reconstructed space with WASD controls!

5. Sample Data & Test Directory

Pre-packaged sample data is placed inside `test/sample_data/`:

- `test/sample_data/frames/`: 30 calibrated multi-view keyframes with ground truth poses.
- `test/bench_poses/`: Verified benchmark comparisons between COLMAP, VGGT, and Postshot.
- Run automated test suite: `python test/test_nav.py`